Mathematical Methods MTH1021 Problem sheet 19

Vector functions. Divergence and curl. Vector identities.

[SUBMIT ANSWERS TO ALL QUESTIONS]

For a vector function (or *vector field*) $\boldsymbol{A}$, the divergence is the scalar $\nabla\cdot\boldsymbol{A}=\dfrac{\partial A_x}{\partial x}+\dfrac{\partial A_y}{\partial y}+\dfrac{\partial A_z}{\partial z}$.

The curl of $\boldsymbol{A}$ is the vector $\nabla\times\boldsymbol{A}=\left(\dfrac{\partial A_z}{\partial y}-\dfrac{\partial A_y}{\partial z}\right)\boldsymbol{i}+\left(\dfrac{\partial A_x}{\partial z}-\dfrac{\partial A_z}{\partial x}\right)\boldsymbol{j}+\left(\dfrac{\partial A_y}{\partial x}-\dfrac{\partial A_x}{\partial y}\right)\boldsymbol{k}$.

Vector identities:

$$\nabla(fg)=\nabla f\,g+f\nabla g, \tag{1}$$
$$\nabla\cdot(f\boldsymbol{A})=\nabla f\cdot\boldsymbol{A}+f\nabla\cdot\boldsymbol{A}, \tag{2}$$
$$\nabla\times(f\boldsymbol{A})=\nabla f\times\boldsymbol{A}+f\nabla\times\boldsymbol{A}, \tag{3}$$
$$\nabla\cdot(\boldsymbol{A}\times\boldsymbol{B})=\boldsymbol{B}\cdot(\nabla\times\boldsymbol{A})-\boldsymbol{A}\cdot(\nabla\times\boldsymbol{B}), \tag{4}$$
$$\nabla\times(\boldsymbol{A}\times\boldsymbol{B})=(\boldsymbol{B}\cdot\nabla)\boldsymbol{A}-\boldsymbol{B}(\nabla\cdot\boldsymbol{A})+\boldsymbol{A}(\nabla\cdot\boldsymbol{B})-(\boldsymbol{A}\cdot\nabla)\boldsymbol{B}. \tag{5}$$

Second-order derivatives:

$$\nabla\cdot\nabla f\equiv\nabla^2 f=\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}+\frac{\partial^2 f}{\partial z^2}, \tag{6}$$
$$\nabla\times\nabla f=0, \tag{7}$$
$$\nabla\cdot\nabla\times\boldsymbol{A}=0, \tag{8}$$
$$\nabla\times(\nabla\times\boldsymbol{A})=\nabla(\nabla\cdot\boldsymbol{A})-\nabla^2\boldsymbol{A}. \tag{9}$$

Vector field $\boldsymbol{A}$ is potential (*irrotational*) if $\nabla\times\boldsymbol{A}=0$; in this case $\boldsymbol{A}=\nabla U$ (U is the potential).
Vector field $\boldsymbol{B}$ is *solenoidal* if $\nabla\cdot\boldsymbol{B}=0$; in this case $\boldsymbol{B}=\nabla\times\boldsymbol{A}$ ($\boldsymbol{A}$ is the vector potential).

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1. Prove the identities involving $\boldsymbol{r}=x\boldsymbol{i}+y\boldsymbol{j}+z\boldsymbol{k}$, $r=\sqrt{x^2+y^2+z^2}$ and constant vector $\boldsymbol{a}$:

   (a) $\nabla r=\boldsymbol{r}/r$, (b) $\nabla\cdot\boldsymbol{r}=3$, (c) $\nabla\times\boldsymbol{r}=0$, (d) $\nabla(\boldsymbol{a}\cdot\boldsymbol{r})=\boldsymbol{a}$.

   [20]

2. Given that $f=x^2y-2z^2+y^3$ and $\boldsymbol{A}=2x\boldsymbol{i}-4xy\boldsymbol{j}+(z+2x)\boldsymbol{k}$, find

   (a) ∇f and $\nabla^2 f$; (b) $\nabla\cdot\boldsymbol{A}$ and $\nabla\times\boldsymbol{A}$.

   [20]

3. Show that $\boldsymbol{A}=(2xy+3)\boldsymbol{i}+(x^2-4z)\boldsymbol{j}-4y\boldsymbol{k}$ is irrotational and find the potential U. After finding U, verify that $\nabla U=\boldsymbol{A}$. [20]

4. Show that the vector field $\boldsymbol{B}=f(r)\boldsymbol{r}$ is solenoidal only if $f(r)=\dfrac{k}{r^3}$ (k is a constant). [20]

   [Hint: find $\nabla\cdot\boldsymbol{B}$ using Eq. (2), set it to zero and solve the differential equation for f.]

5. Show that if the vector field $\boldsymbol{A}=(xyz)^m(x^n\boldsymbol{i}+y^n\boldsymbol{j}+z^n\boldsymbol{k})$ is irrotational (i.e., $\nabla\times\boldsymbol{A}=0$) then either $m=0$ or $n=-1$. [20]

MTH 1021 Problem Sheet 19 Solutions:
Vector functions, Divergence and Curl.
Vector Identities

Q1. $\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$

$r = \sqrt{x^2+y^2+z^2}$

$\underline{a}$ = constant vector

(a) $\underline{\nabla} r = \frac{\partial r}{\partial x}\underline{i} + \frac{\partial r}{\partial y}\underline{j} + \frac{\partial r}{\partial z}\underline{k}$.

and using $r = (x^2+y^2+z^2)^{1/2}$ we find

$$\underline{\nabla} r = \frac{\cancel{2}x}{\cancel{2}\sqrt{x^2+y^2+z^2}}\underline{i} + \frac{\cancel{2}y}{\cancel{2}\sqrt{x^2+y^2+z^2}}\underline{j} + \frac{\cancel{2}z}{\cancel{2}\sqrt{x^2+y^2+z^2}}\underline{k}$$

$$= \frac{x\underline{i} + y\underline{j} + z\underline{k}}{\sqrt{x^2+y^2+z^2}}$$

$$= \frac{\underline{r}}{r}$$

this is a unit vector in the direction of $\underline{r}$.

(b) $\underline{\nabla}\cdot\underline{r} = \nabla\cdot(x\underline{i} + y\underline{j} + z\underline{k})$

$$= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$$

$$= 1 + 1 + 1$$

$$= 3.$$

(c) $\underline{\nabla} \times \underline{r} = \underline{\nabla} \times (x\underline{i} + y\underline{j} + z\underline{k})$

$$= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix}$$

$$= \underline{i}\left[\frac{\partial(z)}{\partial y} - \frac{\partial(y)}{\partial z}\right] - \underline{j}\left[\frac{\partial(z)}{\partial x} - \frac{\partial(x)}{\partial z}\right] + \underline{k}\left[\frac{\partial(y)}{\partial x} - \frac{\partial(x)}{\partial y}\right]\underline{k}$$

$= 0$ since all the partial derivatives above are zero.

(d) $\underline{\nabla}(\underline{a} \cdot \underline{r}) = \underline{\nabla}(a_x x + a_y y + a_z z)$

$$= \frac{\partial}{\partial x}(a_x x + a_y y + a_z z)\underline{i} + \frac{\partial}{\partial y}(a_x x + a_y y + a_z z)\underline{j}$$

$$+ \frac{\partial}{\partial z}(a_x x + a_y y + a_z z)\underline{k}$$

$$= a_x \underline{i} + a_y \underline{j} + a_z \underline{k}$$

$$= \underline{a}.$$

[20 Marks]

Q2.

$$f = x^2 y - 2z^2 + y^3$$

$$\underline{A} = 2x\,\underline{i} - 4xy\,\underline{j} + (z + 2x)\,\underline{k}.$$

(a) $$\underline{\nabla} f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j} + \frac{\partial f}{\partial z}\underline{k}$$

$$= (2xy)\,\underline{i} + (x^2 + 3y^2)\,\underline{j} + (-4z\,\underline{k})$$

$$= 2xy\,\underline{i} + (x^2 + 3y^2)\,\underline{j} - 4z\,\underline{k}.$$

$$\underline{\nabla}^2 f = \underline{\nabla} \cdot \underline{\nabla} f = \frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial y}(x^2 + 3y^2) + \frac{\partial}{\partial z}(-4z)$$

$$= 2y + 6y - 4$$

$$= 8y - 4.$$

(b). $$\underline{\nabla} \cdot \underline{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$= \frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(-4xy) + \frac{\partial}{\partial z}(z + 2x)$$

$$= 2 - 4x + 1$$

$$= 3 - 4x.$$

$$\underline{\nabla} \times \underline{A} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2x & -4xy & z + 2x \end{vmatrix}$$

$$\underline{\nabla} \times \underline{A} = \left[\frac{\partial}{\partial y}(z+2x) - \frac{\partial}{\partial z}(-4xy)\right]\underline{i}$$

$$+\left[\frac{\partial}{\partial z}(2x) - \frac{\partial}{\partial x}(z+2x)\right]\underline{j}$$

$$+\left[\frac{\partial}{\partial x}(-4xy) - \frac{\partial}{\partial y}(2x)\right]\underline{k}$$

$$= (0-0)\underline{i} + (0-2)\underline{j} + (-4y-0)\underline{k}$$

$$= -2\underline{j} - 4y\underline{k}.$$

[20 Marks].

Q3.

$$\underline{A} = (2xy+3)\underline{i} + (x^2-4z)\underline{j} - 4y\underline{k}.$$

Irrotational if $\underline{\nabla} \times \underline{A} = 0$

$$\underline{\nabla} \times \underline{A} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (2xy+3) & (x^2-4z) & -4y \end{vmatrix}$$

$$= \underline{i}\left[\frac{\partial}{\partial y}(-4y) - \frac{\partial}{\partial z}(x^2-4z)\right] - \underline{j}\left[\frac{\partial}{\partial x}(-4y) - \frac{\partial}{\partial z}(2xy+3)\right]$$

$$+ \underline{k}\left[\frac{\partial}{\partial x}(x^2-4z) - \frac{\partial}{\partial y}(2xy+3)\right]$$

$$= \underline{i}\left[-4+4\right] - \underline{j}(0-0) + \underline{k}(2x-2x)$$

$$= 0$$

Hence $\underline{A}$ is irrotational and can be expressed in terms of a potential u by

$\underline{A} = \nabla u$.

To find u we integrate $\int_c \underline{A}\cdot d\underline{r}$ along any curve between an arbitrary point and point (X, Y, Z). lets choose the starting point at the origin and choose the integration path from three segments, parallel to the x, y and z axes.

$(0,0,0) \to (X,0,0)$
$(X,0,0) \to (X,Y,0)$
$(X,Y,0) \to (X,Y,Z)$

then

$$u(X,Y,Z) = \int_c \underline{A}\cdot d\underline{r}$$

$$= \underbrace{\int_{(0,0,0)}^{(X,0,0)} A_x\,dx}_{\substack{\text{on this segment} \\ dy = dz = 0 \\ \text{and } y = z = 0}} + \underbrace{\int_{(X,0,0)}^{(X,Y,0)} A_y\,dy}_{\substack{\text{on this segment} \\ dx = dz = 0 \\ \text{here } x = X \\ \text{and } z = 0}} + \underbrace{\int_{(X,Y,0)}^{(X,Y,Z)} A_z\,dx.}_{\substack{\text{on this segment} \\ dx = dy = 0 \\ x = X \\ y = Y}}$$

$\therefore A_x = 2xy + 3 = 3.$ $\quad A_y = x^2 - 4z = X^2.$ $\quad A_z = -4y = -4Y$

$$= \int_0^X 3\,dx + \int_0^Y X^2\,dy + \int_0^Z (-4Y)\,dz.$$

$$\therefore u(X,Y,Z) = 3X + X^2Y - 4YZ$$

An arbitrary constant can also be added to u since this does not change $\underline{\nabla} u$, so we finally have

$$u(x,y,z) = 3x + x^2y - 4yz + C.$$

Q4.

Using the identity (2)

$$\underline{\nabla} \cdot (f\,\underline{A}) = \underline{\nabla} f \cdot \underline{A} + f\,\underline{\nabla} \cdot \underline{A}$$

for $\underline{B} = f(r)\,\underline{r}$

$$\underline{\nabla} \cdot \left(f(r)\,\underline{r}\right) = \underline{\nabla} f(r) \cdot \underline{r} + f(r)\,\underline{\nabla} \cdot \underline{r}.$$

Now

$$\underline{\nabla} f(r) = \frac{df}{dr}\frac{\underline{r}}{r} \qquad \text{(see lecture notes)}$$

and

$$\underline{\nabla} \cdot \underline{r} = 3 \qquad \text{(see Q1 part (b))}.$$

Hence

$$\underline{\nabla} \cdot \underline{B} = \frac{df}{dr}\frac{\underline{r}}{r} \cdot \underline{r} + 3f(r)$$

Now $\underline{r} \cdot \underline{r} = r^2$

$$\therefore \underline{\nabla} \cdot \underline{B} = r\frac{df}{dr} + 3f(r)$$

For a solenoidal $\underline{B}$, $\underline{\nabla} \cdot \underline{B} = 0$ so we have.

$$r\frac{df}{dr} + 3f = 0$$

$$r\frac{df}{dr} = -3f$$

$$\frac{df}{f} = -3\frac{dr}{r}$$

Integrating both sides

$$\int \frac{df}{f} = -3\int \frac{dr}{r}$$

$\therefore \ln f = -3\ln r + C$

Taking exponents of both sides

$$e^{\ln f} = e^{-3\ln r + C}$$

$$f = e^{c} e^{-3\ln r}$$

renaming $e^{c} = k$ where k is an arbitrary constant

$$f = k\left(e^{\ln r}\right)^{-3}$$

$$= k r^{-3}$$

$$= \frac{k}{r^3} \quad \text{as required}$$

Q5.

$$\underline{A} = (xyz)^m \left(x^n \underline{i} + y^n \underline{j} + z^n \underline{k}\right)$$

$$\underline{\nabla} \times \underline{A} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z. \\ (xyz)^m x^n & (xyz)^m y^n & (xyz)^m z^n \end{vmatrix}$$

$$= \underline{i}\left[\frac{\partial}{\partial y}\left(x^m y^m z^{n+m}\right) - \frac{\partial}{\partial z}\left(x^m y^{n+m} z^m\right)\right]$$

$$-\underline{j}\left[\frac{\partial}{\partial x}\left(x^m y^m z^{n+m}\right) - \frac{\partial}{\partial z}\left(x^{m+n} y^m z^m\right)\right]$$

$$+\underline{k}\left[\frac{\partial}{\partial x}\left(x^m y^{n+m} z^m\right) - \frac{\partial}{\partial y}\left(x^{m+n} y^m z^m\right)\right].$$

$$\underline{\nabla} \times \underline{A} = \underline{i}\left[m x^{m} y^{m-1} z^{n+m} - m x^{m} y^{m+n} z^{m-1}\right]$$
$$+\underline{j}\left[m x^{m+n} y^{m} z^{m-1} - m x^{m-1} y^{m} z^{m+n}\right]$$
$$+\underline{k}\left[m x^{m-1} y^{m+n} z^{m} - m x^{m+n} y^{m-1} z^{m}\right]$$

$$= m x^{m} y^{m} z^{m}\left[\left(\frac{z^{n}}{y} - \frac{y^{n}}{z}\right)\underline{i} + \left(\frac{x^{n}}{z} - \frac{z^{n}}{x}\right)\underline{j} + \left(\frac{y^{n}}{x} - \frac{x^{n}}{y}\right)\underline{k}\right]$$

For $\underline{\nabla} \times \underline{A} = 0$ we can set $\underline{\underline{m = 0}}$ (1st term)

Alternatively we can make the expression in brackets vanish by choosing $\underline{\underline{n = -1}}$, so that

$$\frac{z^{-1}}{y} - \frac{y^{-1}}{z} = \frac{1}{yz} - \frac{1}{yz} = 0$$

$$\frac{x^{-1}}{z} - \frac{z^{-1}}{x} = \frac{1}{xz} - \frac{1}{xz} = 0$$

and $$\frac{y^{-1}}{x} - \frac{x^{-1}}{y} = \frac{1}{xy} - \frac{1}{xy} = 0.$$

and hence $\underline{\nabla} \times \underline{A} = 0$